

preserve organs at risk (OAR) and a plastic cap to stop secondary electrons, while the ^{192}Ir HDR source is positioned in multiple dwell positions close to the treatment surface, producing a uniform dose distribution, high dose rate and steep dose falloff due to the inverse square law. The Papillon 50 (P50) CXRT device was adopted as reference for the applicator design regarding 3 main parameters: radiation field size, dose falloff and treatment time. The applicator design was achieved by Monte Carlo modeling and validated experimentally with film dosimetry.

Results: The slanted edge of the applicator allows the source to be positioned parallel to the treatment surface without excessive source cable curvature (Figure 1-B), enabling a high dose rate for an effective treatment surface of $20 \times 20 \text{ mm}^2$. The Resulting dose falloff is slightly steeper than the one generated by the P50 device using a 22 mm applicator. The relative dose values delivered at depths of 0, 2, 5 and 10 mm are, with normalization at 2 mm depth, respectively, 130, 100, 70 and 43% for the P50 device and 139, 100, 65, and 39% for the rectal applicator. The applicator can deliver an average surface dose of 32 Gy in less than 9 minutes with a 5 Ci source, including the time required to perform channel obstruction verifications before the irradiation.

Conclusions: A rectal applicator for contact radiotherapy was designed to generate dose distributions similar to those of CXRT devices, but using ^{192}Ir HDR brachytherapy sources. The use of multiple channels and TPS integration offers the advantage of increasing the degrees of freedom available to shape the dose distribution and better cover tumors with different shapes e.g. assessed with ultrasound imaging. The novel applicator is able to deliver a uniform dose distribution to a small target volume while the steep dose falloff and lateral shielding protects normal tissues in all other directions.

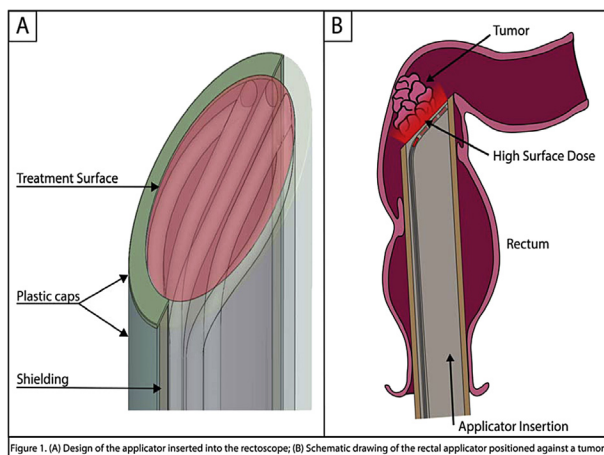


Figure 1. (A) Design of the applicator inserted into the rectoscope; (B) Schematic drawing of the rectal applicator positioned against a tumor

as medium sized tumors with height between 2.5-10mm and diameter less than or equal to 16mm. This study established eye plaque brachytherapy as a standard of care for patients with medium sized choroidal melanoma tumors. There is no current data comparing the efficacy of brachytherapy and enucleation for patients with larger sized tumors. The Purpose of the present study was to use a large, contemporary database to determine current practice patterns and compare survival outcomes between different management options for patients with choroidal melanoma.

Materials and Methods: The National Cancer Database was queried (2004-2014) for histologically-confirmed choroidal melanoma. Inclusion criteria included M0 disease, N0 disease, and complete information regarding tumor size and vital status. The Chi-Squared test and Fisher exact tests were used to compare categorized demographic and clinical variables in both arms. Kaplan-Meier analysis evaluated overall survival (OS). Cox proportional hazards analysis determined variables associated with OS. Patients were divided into cohorts representing small, medium, and large tumors based on COMS trial size criteria.

Results: A total of 3,777 patients met the selection criteria; 2964 (78%) patients received brachytherapy and 813 (22%) of patients were treated with enucleation. Median follow-up was 38 and 30 months for brachytherapy and enucleation, respectively. Brachytherapy patients were more likely to be from educated zip codes, have private insurance, and live further away from where they were treated. Enucleation patients were more likely to be treated at an academic center and be uninsured. Amongst all patients, median overall survival was 133 vs. 65 months for brachytherapy and enucleation, respectively ($p < 0.001$). Median survival for COMS small tumors was 132 vs. 96 months, for COMS medium tumors was 131 vs. 69 months, and for COMS large tumors was 130 vs. 59 months for brachytherapy and enucleation, respectively ($p < 0.001$). Multivariate Cox regression found older age (hazard ratio [HR] = 1.43, 95% confidence interval [CI] = 1.20-1.85), more comorbidities (HR = 1.56, 95% CI = 1.29-1.87), and increasing COMS size (HR = 1.19, 95% CI = 1.06-1.34) were negative prognosticators of survival. Brachytherapy was a positive prognosticator of survival (HR = 0.36, 95% CI = 0.31-0.43).

Conclusions: Patients selected for brachytherapy had improved survival compared to enucleation in all COMS size cohorts in this large retrospective review. Brachytherapy is a reasonable treatment option for certain patients with COMS large size tumors. The retrospective nature of this analysis is recognized as a limitation. Further prospective studies are warranted to confirm these findings.

MSOR07 Presentation Time: 5:45 PM

Clinical Efficacy and Prognostic Factors of CT-Guided ^{125}I Brachytherapy for the Treatment of Retroperitoneal Metastatic Lymph Nodes

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Purpose: Due to the unique anatomical location of retroperitoneal metastatic lymph nodes, current treatment options are limited. This study was designed to explore the clinical efficacy and prognostic factors of CT-guided ^{125}I brachytherapy for the treatment of retroperitoneal metastatic lymph nodes.

Materials and Methods: We retrospectively evaluated 92 patients who received ^{125}I brachytherapy for retroperitoneal metastatic lymph nodes from April 24, 2012 to August 4, 2016. The main study index was local tumor progression-free survival (LTPFS). A layered Cox proportional hazards model was established to filter out the independent factors affecting LTPFS.

Results: The median LTPFS was 8 (1-87) months and the average LTPFS was 15.2 months. Six months after ^{125}I brachytherapy, 27 patients (29.3%) achieved complete remission (CR). Metastatic lymph node with previous radiotherapy ($p=0.017$), uniform density ($p=0.009$), clear boundaries ($p=0.011$), regular morphology ($p < 0.001$), and < 3 organs at risk of metastasis ($p=0.020$) were associated with better LTPFS. Necrotic lymph nodes ($p < 0.001$), fusion ($p=0.003$), and invasion of vessels visible on images ($p < 0.001$) were associated with poor LTPFS. Puncture path through abdominal wall or paravertebral approach were also associated

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Eye Plaque Brachytherapy versus Enucleation for Ocular Melanoma: An Analysis from the National Cancer Database

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Purpose: The Collaborative Ocular Melanoma Study (COMS) group conducted a Phase III randomized trial that demonstrated medical equipoise between patients with choroidal melanoma treated with either enucleation or eye plaque brachytherapy, with no survival differences found between the treatment cohorts. The inclusion criteria were defined